Q U A R K - H A W K I N G B R I D G E

The Quark-Hawking Bridge. Volume I: The Theorems

thirteen theorems T1–T13 on the gap, the door, the trap and the barrier

Isaev, Ishak Khamzatovich (Nalchik, KBR, RF)
co-author: wild8highlander (GitHub)
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1. The riddle and the proposal

Quarks are the fundamental constituents of protons and neutrons, and they are never observed alone — only inside hadrons. The phenomenon is confinement. Quantum chromodynamics supplies overwhelming lattice evidence, but no analytic proof exists that quarks cannot live in isolation; the statement is part of the Yang-Mills mass-gap millennium problem.

This volume fixes the theoretical layer of the Quark-Hawking Bridge package — a research program that inherited its honesty discipline from the Collapse-Hawking Bridge (CHB v2.1.0). The proposal is a four-link mechanism. H1: quark triplets are welded at maximal density — the critical-collapse regime of CHB, where the born mass tends to zero and the mean horizon density tends to the Planck density. H2: the three dynamic corrections of the septinial family — the Berry quanta and the braking quanta of the pi/2, pi/3, pi/7 sectors — are not decoration but a DRIVE: they vibrate, shift the discretized geometry and open a real spectral gap. H3: Hawking pairs leave the horizon, the weak channel violates parity (Wu et al. 1957), and a residual matter imbalance survives annihilation — the Wu door. H4: a triplet bound at maximal density cannot shed a member — the unsolved three-body problem shares energy chaotically, and the gap stands as an oscillating geometric wall: bound chaos.

The status is fixed honestly: **the confinement mechanism of the package is a marked HYPOTHESIS**. Every theorem of this volume is a machine-checked fact about the committed toy models, not a derivation of QCD. Since v1.2.0 the proof / measurement / comparison / hypothesis separation is machine-enforced: the Evidence Engine claims registry (27 claims, five statuses, uncertainty budgets, falsification criteria) is pinned to the package build and summarised in section 17.

2. Notation and units

The core of the package speaks in code units: the hopping is tied to the grid step by $J = 1/h^2$, the drive base frequency is Omega_0 = 2, the drive tones are (2, 3, 7)·Omega_0, the Floquet period is T_0 = pi. The physical layer (Schwarzschild thermodynamics, the contact points) is expressed in SI through CODATA. The single source of randomness of the whole package is numpy.default_rng(20261010); re-running any campaign reproduces its verdict bit-for-bit.

The committed G-model geometry: a Dirichlet chain of N = 64 sites carrying the parabolic well $q_n = (\kappa/2)x_n^2 - q_{pk}$ with kappa = 0.05 and q_pk = 0.4 (a shallow well with exactly two levels), the truncation length $L = \sqrt{8 \cdot q_{pk}/\kappa}$ — the well ends where the potential ends. The vortex layer of C11 speaks in circulation code units: lengths in units of the triangle side a, time in units of a^2/Γ.

3. T1 — the six correction quanta (exact)

The spinor phases of the Klein-quartic family are delta_A = pi/2, delta_B = pi/3, delta_C = pi/7 (the septinial sector). The quanta follow the exact maps beta(d) = $d^2/2$ (Berry quanta) and alpha(d) = $d^5/22$ (braking quanta, k = b2(K3) = 22), inherited from the spinor_corrections block of the parent repository:

bA = pi^2/8 = 1.2337005501361697 aA = pi^5/704 = 0.43468705225182025
bB = pi^2/18 = 0.5483113556160755 aB = pi^5/5346 = 0.057242739391186152
bC = pi^2/98 = 0.10071024899070774 aC = pi^5/369754 = 0.0008276304915843546

Status: exact identities; the 50-digit layer reproduces all six to a relative deviation of at most 7.8e-17 (qhb.validate, check 1; IVP04 — the independent derivation). In this package the six are the drive: the Berry quanta modulate the geometry transversally, the braking quanta longitudinally.

4. T2 — the selection rules (exact + machine)

> **Statement T2.1 (exact).** The gap is strictly even under the global Berry flip b → −b: U(−b, a) = sigma_z U(b, a) sigma_z, the eigenphases coincide. Machine check: deviation exactly 0.0. Campaign C10 (v1.2.0) generalises the rule: the evenness holds on EVERY configuration with a nonempty transverse part — all seven checks return machine zeros.
>
> **Statement T2.2 (exact).** A purely longitudinal drive (v = 0, braking only) is invisible to the gap at all orders: over the integer tones the cosine integral vanishes and U equals the unperturbed one. Machine check: deviation 8.9e-16 (machine zero); C10 confirms the zero response at doubled Floquet resolution (IVP10). Physical reading: braking without vibration opens nothing — the braking quanta act only jointly with the Berry drive, through the Magnus commutator [w·sigma_z, v·sigma_x].
>
> **Statement T2.3 (data, not smoothed).** Flipping a alone is not an exact symmetry: the residual 1.5e-6 at beta = 0.4 is the genuine odd-in-a second-order component (the a·b cross terms). It is carried in the verdict as data.

Table T2 — the machine checks of the selection rules (c1_verdict.json, c10_verdict.json)

check	deviation	status
b → −b (global)	0.0	exact
a-only, all orders	8.9e-16	machine zero
a → −a (with v ≠ 0)	1.5e-6	data (ab-cross)
b = 0 identity	0.0	exact

check	deviation	status
b → −b on all 7 transverse configs (C10)	0.0	exact

5. T3 — the gap law (window [0, 0.2])

> **Statement T3.** On the committed window beta in [0, 0.2] (21 points) the Floquet gap obeys the quadratic law Delta_F = Delta_0 + B2·beta^2 with the committed coefficients:

Delta_0 = 0.523004556028 B2 = -5.49670294e-2 r^2 = 0.99999982
max|residual| = 4.7e-7

The sign B2 < 0: the drive CLOSES the gap — the dynamic corrections shift the geometry, exactly as the hypothesis proposes. The sector structure of the septinial family is visible in the response: |B2_A(pi/2)| = 5.05483488e-2, |B2_B(pi/3)| = 4.40077100e-3, |B2_C(pi/7)| = 2.71009207e-5 — the hierarchy of roughly 1865 : 1. Beyond the window the response becomes quartic-dominated; the wing [0.25, 0.8] is carried unfitted (the honest pre-asymptotic story). The cross-term share between the sectors is 1.3e-4: the tones are separated, and on the committed window the trio acts as one dynamic object with one coefficient. The causal map of C10 (section 15) confirms the additivity and the hierarchy by the factorial ablation.

6. T4 — the Gascheau criterion (exact negative)

> **Statement T4 (classical, exact).** The equilateral Lagrange solution of the three-body problem is linearly UNSTABLE for equal masses. The Gascheau (1843) criterion: stability requires nu = (m1m2 + m2m3 + m3m1)/(sum m)² < 1/27, while equal masses give nu = 1/3. The characteristic equation lambda^4 + lambda^2 + (27/4)nu = 0 has complex roots with the growth rate sqrt(3/2) ≈ 1.2247 per mean motion.

Symbolic verification (IVP06, SymPy): the stability condition 1 − 27nu ≥ 0 is checked in exact rationals; for nu = 1/3 one gets 1 − 9 < 0. Status: UNSTABLE — exact. Physical reading: no static rigid triangle exists; the naive picture of three quarks rigidly sitting at the vertices dies on this exact negative. Rigidity, if any, must come from outside — from the dynamic wall of T6, from the invariants of T12, or from the interaction structure itself (T11).

7. T5 — the free trio unbinds (negative)

> **Statement T5 (negative, measured).** On the committed weakly-bound ensemble (E in [−0.6, −0.05), N = 300, seed 20261010) the Newtonian trio without the wall unbinds:

> P_esc = 0.2433 ± 0.0248. Static gravity does not confine. At packaging 18187 draws were rejected (honest bookkeeping); the survivor energy drift is 2.0e-5 (relative, on the wall-free runs).

This negative is one half of the mechanism. Without the wall the chaos of the three-body problem CHANNELS energy into escape: a quarter of the weakly-bound trios lose a member within the committed time. Confinement needs the second ingredient — the gap of T3 standing as a wall. The exact stratification introduced in v1.2.0 (T13, section 15) now separates three regimes: exact impossibility, non-observation on the horizon, and finite-time sealing.

8. T6 — the wall seals; bound chaos

> **Statement T6 (law, measured).** The oscillating geometric wall $U(r, t) = V0 \cdot (1 + \cos(\Omega \cdot t)) \cdot \max(0, (r - R_b)/w)^2$ with $R_b = 8$, $w = 1$, $p = 2$ standing at the committed gap scale seals the system: at every committed $V0 \geq 0.4$ cell all 36 grid cells ($V0 \times \Omega$, Ω in $\{0, 1, 2, 4\}$) give $P_{esc} < 1e\text{-}4$. The $P_{esc} = 1/e$ boundary lies between the first non-zero cell and zero for every Ω.

The key diagnostic is the Benettin Lyapunov exponents: free median 0.2615, trapped 0.2885. Both positive and comparable: the wall does not change the chaos — it changes the EXIT. The dynamics inside stays chaotic (the unsolvability of the three-body problem has not gone anywhere), but the energy cannot collect into an escape channel against the oscillating wall. Bound chaos is the committed confinement picture of the toy model. Honest caveats: the softening $\text{eps}^2 = 1e\text{-}2$ is a committed regularization (the observables are escape statistics, not microdynamics); for $V0 > 0$ the wall does work and the energy-drift metric is not a quality metric; since v1.2.0 the sealing statement carries the exact T13 stratification — the static seal is exact, the oscillating one is finite-time.

9. T7 — the Wu door (closed form)

> **Statement T7 (exact in the toy model).** The linear two-rate model: two species born at the Hawking rate Γ_b, annihilating at $\Gamma(1 -/+ \alpha)$, freeze-out at $t_f = 1/H$, $x = \Gamma/H$. The residual imbalance is an exact rational function:

$$f(u) = (1 - e^{-x\,u})/u$$
$$r(\alpha, x) = [f(1-\alpha) - f(1+\alpha)] \,/\, [f(1-\alpha) + f(1+\alpha)]$$

Limits (the 50-digit layer): $r(0, x) = 0$; $r(\alpha, \infty) = \alpha$; $r(\alpha, 0) = 0$. The seeded stochastic realisation (Poisson births, thinned survival; 200,000 realisations per cell, 30 cells) agrees with the formula. At the Wu anchor ($\alpha = 0.9$ — an approximate literature anchor, marked) and $x = 100$: $r = 0.899996$ — the door opens almost fully.

The honest scale. To leave only the observed baryon-to-photon ratio eta = 6.1e-10 the freeze-out must sit at x = 1.36e-9 — practically immediately after the door opens. The model realises the Sakharov ingredients qualitatively (CP violation — the alpha channel; departure from equilibrium — the Hawking flux and the freeze-out), but NOT baryogenesis: the baryon number is conserved in the toy. The door is a door.

10. T8 — the QCD contact (closed form)

> **Statement T8 (exact arithmetic on the anchors).** The mass of the black hole whose Hawking temperature equals the lattice-QCD deconfinement temperature of 155.16 MeV:

M_QCD = hbar c^3 / (8 pi G kB T_QCD) = 6.814006e10 kg

Literature comparison: the primordial holes of the QCD epoch are ~1e11 kg — the same order (ratio 0.68), a comparison only, no derivation — the 0.67%-gap honesty of CHB. The second contact is inherited: T_H(M) = T_CMB gives 4.5016e22 kg. The bridge master table carries both points: T_H/T_QCD and T_H/T_CMB in each of the 15 rows (3 members of the inherited f~ ensemble × 5 input scales).

11. T9 — the spectral floor (the detector)

> **Statement T9 (the detector law).** A gap that cannot be measured does not exist for the detector. The committed period-sampled Floquet spectroscopy (the exact period propagator, one sample per period, FFT windows with quadratic peak interpolation) resolves the beat only if (i) at least three beats fit the window and (ii) the peak error is under half the Fourier resolution 2/N.

Committed: T_sim = 20 — UNRESOLVED (1.53 beats < 3 — the honest-negative language of CHB); T_sim = 40…320 — resolved, the errors 4.6e-3 → 3.2e-5 ride under the 2/N envelope (the measured exponent 2.39 is interpolation noise below the envelope; carried as data). The floor is a property of the DETECTOR, not of the physics. Campaign C9 (v1.2.0) turns exactly this envelope into the phase threshold: the map cells are classified against 2/N.

12. T10 — the gap-drift law (the heir)

> **Statement T10 (a measured law with an exact channel).** The sampled geometry shifts the gap. Lemma A (inherited from CHB T10, exact): a parabola sampled with step h loses kappa·h²/8 of its peak in the worst alignment; the committed centred chain with even N IS the worst alignment — the machine check gives exactly 0.0. With the measured response s = d(Delta_0)/d(q_pk) = −1.00398294:

Delta_cont - Delta_N = C h^2 + O(h^3), C_measured = +3.0089e-2
C_exact = s*kappa/8 = -6.275e-3, share = -0.209 (opposing, signed)
p_eff = 1.948 -> 1.974 -> 1.987

Two honest negatives travel with the theorem: Delta_cont is NOT claimed (the two-parameter fit 0.523460440588 is displayed without a claim — the same rejection CHB applied to A*_inf), and the two-channel decomposition of the remainder is a marked hypothesis. The sign mirror: the CHB threshold ladder approached from above, the gap ladder approaches from below; the sign is carried, not hidden. The full derivation lives in docs/gap_law.md of the package.

13. T11 — the stability contrast (exact; v1.2.0)

> **Statement T11 (exact, classical + machine).** The equilateral triangle of three identical same-sign vortices (the Kirchhoff problem, 1876) is a rigidly rotating relative equilibrium with omega = Gamma_tot/(2 pi a²) = 3/(2 pi) = 0.477464829… (unit circulations, unit side), and its corotating-frame linearization is NEUTRAL: the exact analytic Jacobian has the spectrum {0, 0, +i omega, −i omega, +i omega, −i omega} — every eigenvalue on the imaginary axis or at the origin.

The symbolic proof (IVP11, SymPy): the characteristic polynomial of the corotating linearization, computed in exact arithmetic over Q(sqrt(3), pi), is IDENTICALLY lambda²·(lambda² + omega²)² — the deviation is exactly 0. The machine layer (C11, block B1): the exact Jacobian (no finite differences) gives max|Re| = 2.8e-17 on the imaginary pairs and 4.7e-9 of solver noise on the genuine double-zero eigenvalue; the polynomial identity is reproduced numerically with dev 1.75e-16.

The measured complement (C11, block B2): three full rotations of the triangle are rigid to 1.5e-14 (the three pairwise distances constant), all four classical invariants I, H, P, Q drift by less than 2.2e-14, and the measured rotation rate matches the exact one to 1.6e-14.

Table T11 — the contrast of the SAME triangle (c2_verdict.json, c11_verdict.json)

system	interaction	the equilateral triangle	status
three-body problem (Newton)	1/r attraction	unstable (Gascheau): growth sqrt(3/2)	exact negative (T4)
three-vortex problem (Kirchhoff)	logarithmic, advection	neutrally stable: {0, 0, ±i omega, ±i omega}	exact (T11)

The reading for link H4: "three-ness" by itself does not create the escape channel — the INTERACTION decides. The trap is a property of the physics, not of the triangle. Status: an exact statement about two committed toy models; its projection on QCD remains the marked hypothesis H4.

14. T12 — the invariant wall (measured; v1.2.0)

> **Statement T12 (measured on the classical manifold).** The mixed-sign trio (1, 1, −1) launched from the classical right-isosceles configuration r1 = sqrt(2)·xhat, r2 = sqrt(2)·yhat, r3 = r1 + r2 annihilates ALL FOUR classical invariants of the three-vortex problem at once: I = sum Gamma|r|² = 0, H = 0, P = 0, Q = 0 — machine zeros at 50 digits (mpmath; IVP11). This is the classical Aref (1979) collapse manifold; the angular impulse I is the classical prototype of the Chaplygin topological integral.

On the committed horizon t = 12 the trio evolves NON-rigidly (the separations grow from (2, sqrt(2), sqrt(2)) and stay bounded: the maximum distance 2.7654) while all four invariants stay pinned to machine precision (drift ≤ 1.5e-12): **the system is bound WITHOUT ANY WALL** — confinement by invariants, the topological complement of the geometric wall of T6.

Honest caveats: the manifold I = H = P = Q = 0 is measure-zero in phase space; the self-similar collapse separatrix r ∝ (t_c − t)^(1/2) — a classical result (Aref 1979) — is NOT integrated by the package; the boundedness is measured on the committed horizon, not proven for all times. The implementation follows the classical Kirchhoff equations with the package's own code paths; the invariants are re-derived and re-measured, never imported.

15. T13 — the static barrier (exact; v1.2.0)

> **Statement T13 (exact for the committed engine).** With the wall frozen (Omega = 0) the total energy of the engine is conserved and the wall potential grows without bound: U(r) = 2·V0·s(r)², s = max(0, (r − R_b)/w). Every solution of the committed softened engine obeys the exact radius bound

$$r_i(t) <= R_b + w * sqrt((E0 + 3/sqrt(soft2)) / (2 V0))$$

(the softened pair floor −3/sqrt(eps²) carried exactly; the derivation is one line: K ≥ 0 and the remaining terms are bounded below, hence U_wall(r_i) ≤ E0 + 3/sqrt(eps²)). Escape — crossing R_ESC = 12 — is therefore IMPOSSIBLE whenever

$$V0 > V* = (E0_max + 3/sqrt(soft2)) * w^2 / (2 (R_ESC - R_b)^2) = 0.9359375$$

on the committed ensemble band. The C2 cells with V0 ≥ 1.6 at Omega = 0 are sealed EXACTLY — a theorem about the engine equations, not a Monte-Carlo frequency. Machines: 0/300 escapes measured on the three cells (V0 = 0.4, 0.8, 1.6; Omega = 0) with bound margins ≥ 2.76 and the energy conserved to 7.6e-5 absolute.

Table T13 — the honest stratification of the sealing (c11_verdict.json)

regime	statement	status
V0 ≥ V*, Omega = 0	escape impossible (the radius bound)	exact, proven
V0 < V*, Omega = 0	escape allowed by the bound, NOT OBSERVED (0/300, T = 12)	measured on the horizon
Omega > 0	the wall does work: median |dE| = 0.146; the C2/C7 sealing is finite-time	measured, honest complement

The honest caveat: the theorem is stated for the COMMITTED SOFTENED engine (the pair floor enters the bound); at Omega > 0 the wall pumps energy, conservation fails, and the verdict stays finite-time — both marks travel with the numbers.

16. The verification matrix

Eleven campaigns C1–C11 produce eleven committed deterministic verdicts; the hostile audit IVP-11 (eleven checks from sources of truth OUTSIDE the committed results) confirms them 11/11 PASS; 94 pytest pins bind the numbers to the build. The matrix below is the summary; the verdict addresses live in docs/validation.md of the package.

Table 16.1 — the campaigns and their theorems

campaign	content	theorems	runtime
C1	statics, the gap law, sectors, T2	T1–T3	~1 s
C2	Gascheau, the trap grid, Lyapunov	T4–T6	~3.5 min
C3	the Wu door: the 6x5 grid x 200k	T7	~1 s
C4	the detector ladder T_sim 20..320	T9	~1 s
C5	the drift ladder N 32..512, channels	T10	~1 s
C6	the three-branch physical mapping	T8 (contact)	~1 s
C7	the sigma ladder, the V0* collapse	T6 (ext.)	~5 min
C8	the pairwise door, RK4 ladder, mpmath	T7 (negative)	~10 s
C9	the phase map, the valley, the counterexample	T3 (scope)	~11 s

campaign	content	theorems	runtime
C10	the 8-config ablation, forecasts	T2/T3 (cause)	~4 s
C11	the vortex anchor, Aref, the static barrier	T11–T13	~35 s

Table 16.2 — the audit IVP-11 (11/11 PASS)

IVP	check	verdict
IVP01	release integrity (verdicts + sha256)	PASS
IVP02	the canonical job (fresh interpreter)	PASS
IVP03	the dependency audit (ast-scan)	PASS
IVP04	the 50-digit derivation of closed forms	PASS
IVP05	the dimensions gates (SI)	PASS
IVP06	the symbolic layer (Gascheau + Lemma A)	PASS
IVP07	the convergence orders (p_eff)	PASS
IVP08	the seed determinism (bit-identical MC)	PASS
IVP09	the phase-map gate (own bisection)	PASS
IVP10	the competing-model gates (M = 3840)	PASS
IVP11	the vortex layer (SymPy charpoly, 50-digit Aref)	PASS

17. The honesty register (the claims registry)

Since v1.2.0 the honesty register is machine-supported: results/claims_registry.json lists the 27 load-bearing statements of the package, each with a status from the five-element enum (exact / measured-law / honest-negative / hypothesis / comparison), a verdict address, a five-field uncertainty budget (statistical / discretization / solver / parameter / model) and a pre-committed falsification criterion. The qhb.claims validator checks the schema, the addresses and the budgets on every build; the tests break if the registry breaks.

Table 17.1 — the registry statuses (v1.2.0)

status	count	example
exact	5	the T11 spectrum {0, 0, ±i omega, ±i omega}; the T13 barrier V*

status	count	example
measured-law	12	the gap law T3; the C9 narrowing valley
honest-negative	7	the free-trio P_esc; Delta_cont not claimed
hypothesis	1	the confinement mechanism (furnace-gap-door-trap)
comparison	2	the QCD contact 0.68x; the window Lambda/E_gap = 4.09

The registry rule: a conclusion can never be marked exact because one error component is zero — the model-uncertainty field of every toy-model fact points at the toy itself. Quoting a number of the package obliges carrying its status and its verdict address (the honesty discipline is part of the license).

18. References

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